

FIRST PROOF • SECOND BATCH PROBLEM SUBMISSION

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PROBLEM STATEMENT.

Let T denote a finite, weighted tree with vertex set V , edge set E , and weight function $w : V \rightarrow \mathbb{Z}$. Let $L(T)$ denote the free abelian group generated by V , equipped with the symmetric, bilinear form $\cdot$ whose value on a pair of generators $u, v \in V$ is given by

$$u \cdot v = \begin{cases} w(u), & u = v; \\ -1, & (u, v) \in E; \text{ and} \\ 0, & u \neq v, (u, v) \notin E. \end{cases}$$

Suppose that the form is positive definite, and suppose that there exists a single vertex v for which the vertex weight $w(v)$ is less than the vertex degree $d(v)$. Prove that T contains a vertex u such that u is an irreducible element of $L(T)$, i.e. there do not exist nonzero lattice elements $a, b \in L(T)$ such that $u = a + b$ and $a \cdot b \geq 0$.

1. CONTEXT

A finitely generated, free abelian group equipped with a real-valued, symmetric bilinear form is called a *lattice*. It is *integral* if the form takes integer values, *rational* if the form takes rational values, and *positive definite* if the form is positive definite. An element of a positive definite lattice L is called *reducible* if it is possible to express it as a sum of nonzero elements $a, b \in L$ such that $a \cdot b \geq 0$, and it is called *irreducible* otherwise. For example, an element $a \in L$ of minimal nonzero self-pairing $|a|^2 = a \cdot a$ in a lattice L is irreducible, such as an element of self-pairing 1 in an integral lattice, and the irreducible elements in the lattice of integer points in Euclidean space are precisely the elements of self-pairing 1. Irreducible lattice elements specify the facets of the Voronoi polyhedron of L . The lattices $L(T)$ considered in the problem statement are called *tree lattices*. A vertex v in a weighted tree is *bad* if $w(v) < d(v)$. In this language, the problem statement is to prove:

Theorem 1.1. *If a weighted tree T contains a single bad vertex and $L(T)$ is positive definite, then $L(T)$ contains an irreducible vertex element.*

Theorem 1.1 is motivated by a conjecture of Walter Neumann and Don Zagier [NZ85, Neu89]:

Conjecture 1.2. *If T is a weighted tree and $L(T)$ is isometric to the lattice of integer points in Euclidean space, then there exists a vertex $v \in V$ with weight $w(v) = 1$.*

Conjecture 1.2 stems from a problem in low-dimensional topology concerning integer homology cobordism and gauge theory. The original version of Conjecture 1.2 in [NZ85] is posed as a question for star-shaped trees, i.e. those containing a single vertex v with $d(v) \geq 3$. Issa and McCoy settled this conjecture in [IM20]. Neumann posed the more general Conjecture 1.2 in [Neu89], where he attributed it to himself and Zagier.

Generalizing [IM20], we proved Conjecture 1.2 under the assumption that T has a single bad vertex (compare [OS03, Definition 1.1]). We then realized that we could prove the more general Theorem 1.1. It leads us to conjecture the following improvement on Conjecture 1.2:

Conjecture 1.3. *If T is a weighted tree and $L(T)$ is positive definite, then at least one vertex of T is irreducible in $L(T)$.*

We have two primary motivations for proposing this problem. Firstly, we believe that this problem has several characteristics making it a good AI test case. Although Theorem 1.1 can be proven with relatively little background, our proof involves several subtle steps. Furthermore, there is very little literature relating to Conjecture 1.2, and our proof of Theorem 1.1 does not resemble the techniques used in [IM20]. Secondly, we are simply curious to see whether an AI attempt could shed fresh light on Conjectures 1.2 and 1.3.

2. PREPARATION AND LEMMAS.

2.1. Cut and flow lattices. We review some basic definitions about cut and flow lattices on graphs [GR01, Gre13]. Let G be a finite, loopless graph with vertex set $V = V(G)$ and edge set $E = E(G)$, possibly with parallel edges. We use $e(v, w)$ to denote the number of edges between a pair of distinct vertices v and w in G .

Choosing an arbitrary orientation for the edges of G endows G with the structure of a CW-complex. This allows us to consider the integral cellular chain complex

$$(2.1) \quad 0 \longrightarrow C_1(G) \xrightarrow{\partial} C_0(G) \longrightarrow 0,$$

where $\partial(e) = v - w$ for an edge e oriented from w to v . We equip $C_1(G)$ and $C_0(G)$ with bilinear pairings such that E and V are orthonormal bases, respectively. These pairings give rise to the adjoint map

$$(2.2) \quad \begin{aligned} \partial^* : C_0(G) &\longrightarrow C_1(G), \\ v &\longmapsto \sum_{e \in E} (\partial(e) \cdot v) e. \end{aligned}$$

The *cut lattice* on G is defined to be

$$\text{Cut}(G) = \text{im } \partial^* \subseteq C_1(G).$$

Clearly the images of the vertices span $\text{Cut}(G)$ and for any pair of vertices, $u, v \in V$, we have

$$(2.3) \quad \partial^* u \cdot \partial^* v = \begin{cases} d(u) & \text{if } u = v \\ -e(u, v) & \text{if } u \neq v. \end{cases}$$

The *flow lattice* of G is defined to be

$$\text{Flow}(G) = \ker \partial \subseteq C_1(G).$$

Since ∂^* is adjoint to ∂ , the lattices $\text{Cut}(G)$ and $\text{Flow}(G)$ are orthogonal to one another in $C_1(G)$.

If L is an integral lattice, then $L \otimes \mathbb{Q}$ is a vector space with an induced symmetric, bilinear form. Within it sits the dual lattice $L^* = \{\lambda \in L \otimes \mathbb{Q} : \lambda \cdot v \in \mathbb{Z}, \forall v \in L\}$. A sublattice $L' \subset L$ is *primitive* if it satisfies any of the equivalent conditions:

- $L \cap (L' \otimes \mathbb{Q}) = L'$, the intersection taking place within $L \otimes \mathbb{Q}$;
- L' admits a basis which completes to one of L ; and
- the restriction/projection map $L^* \rightarrow (L')^*$ surjects.

The lattice $C_1(G)$ is self-dual, and $\text{Cut}(G)$ and $\text{Flow}(G)$ are both primitive sublattices of $C_1(G)$ [Gre13, Proposition 3.1]. The inclusion of the orthogonal direct sum $\text{Cut}(G) \oplus \text{Flow}(G) \hookrightarrow C_1(G)$ induces an inclusion of dual lattices $C_1(G) \hookrightarrow \text{Cut}(G)^* \oplus \text{Flow}(G)^*$. We uniquely express every $x \in C_1(G)$ in the form $x = x_C + x_F$ with respect to this decomposition.

2.2. The projection length formula. The result of this section is a formula for the self-pairing of the projection $x_C \in \text{Cut}(G)^*$ of an element $x \in C_1(G)$.

Let G be a connected graph with a choice of orientation on its edges. Let (A, B) be an ordered partition of the vertex set of G . The cut vector defined by (A, B) is

$$\chi_{A,B} = \sum_{e \in E(A,B)} \varepsilon_{A,B,e} e \in C_1(G),$$

where

$$\varepsilon_{A,B,e} = \begin{cases} 1 & \text{if } e \text{ oriented from } A \text{ to } B \\ -1 & \text{if } e \text{ oriented from } B \text{ to } A. \end{cases}$$

Evidently, we have that $\chi_{A,B} = -\chi_{B,A}$. Observe also that $\chi_{A,B}$ can be expressed as

$$(2.4) \quad \chi_{A,B} = - \sum_{v \in A} \partial^* v.$$

This implies that $\chi_{A,B} \in \text{Cut}(G)$.

A two-component spanning forest of G is an unordered pair $\{A, B\}$ of subgraphs of G such that A and B are trees and A and B partition the vertices of G . Given an element $x \in C_1(G; \mathbb{R})$, we define its square-flux on $\{A, B\}$ to be

$$\mathcal{F}_{\{A,B\}}(x) = |\chi_{A,B} \cdot x|^2 = |\chi_{B,A} \cdot x|^2.$$

Explicitly, if x takes the form $x = \sum_e x_e e$, this is

$$(2.5) \quad \mathcal{F}_{\{A,B\}}(x) = \sum_{e \in E(A,B)} x_e^2 + \sum_{\substack{e, e' \in E(A,B) \\ e \neq e'}} 2\varepsilon_{A,B,e} \varepsilon_{A,B,e'} x_e x_{e'}.$$

Lemma 2.1. *Let G be a connected graph and let $x \in C_1(G)$. If $x = x_C + x_F$, where $x_C \in \text{Cut}(G)^*$ and $x_F \in \text{Flow}(G)^*$, then*

$$|x_C|^2 = \frac{1}{\tau} \sum_{\{A,B\}} \mathcal{F}_{\{A,B\}}(x),$$

where the sum is over all two-component spanning forests and τ is the number of spanning trees in G .

Remark. One can deduce Lemma 2.1 from an expression for x_C due to Kirchhoff [Kir47]. (For a modern treatment, see [GR01, Theorem 14.8.1].) We give a self-contained argument in the spirit of “completing the square.”

Proof. A *connected spanning unicyclic subgraph* R of G is a subgraph which is connected, which contains every vertex of G , and which contains a unique cycle C_R . Deleting any edge of this cycle therefore results in a spanning tree for G . Given a connected spanning unicyclic subgraph R , we define a circulation vector for R to be a vector of the form $\omega_R = \sum_{e \in C_R} \delta_{R,e} e$, where the $\delta_{R,e} \in \{\pm 1\}$ are chosen so that $\partial \omega_R = 0$. Note that the circulation vector is unique up to an overall change of sign. Given an element $x \in C_1(G; \mathbb{R})$, we define its circulation on R to be

$$\mathcal{C}_R(x) = |\omega_R \cdot x|^2.$$

The circulation can be calculated by the formula

$$(2.6) \quad \mathcal{C}_R(x) = \sum_{e \in C_R} x_e^2 + \sum_{\substack{e, e' \in C_R \\ e \neq e'}} 2\delta_{R,e}\delta_{R,e'} x_e x_{e'}.$$

We notice a number of natural bijections:

- Given a fixed edge e , the set of two-component spanning forests $\{A, B\}$ with $e \in E(A, B)$ is in bijection with the set of spanning trees containing e .
- The set of connected spanning unicyclic subgraphs R with $e \in C_R$ is in bijection with the set of spanning trees not containing e .
- Given a pair of distinct edges e and e' , the set of two-component spanning forests $\{A, B\}$ with $e, e' \in E(A, B)$ is in bijection with connected spanning unicyclic subgraphs R such that $e, e' \in C_R$. We obtain $\{A, B\}$ by deleting e and e' from R . Moreover note that under this bijection we always have $\varepsilon_{A,B,e}\varepsilon_{A,B,e'} = -\delta_{R,e}\delta_{R,e'}$.

Thus summing (2.5) and (2.6) over all two-component spanning forests and all connected spanning unicyclic subgraphs we obtain

$$(2.7) \quad \tau |x|^2 = \sum_{\{A,B\}} \mathcal{F}_{\{A,B\}}(x) + \sum_R \mathcal{C}_R(x).$$

Since x_C is orthogonal to every element of $\text{Flow}(G)$, we have $\mathcal{C}_R(x_C) = 0$ for every connected spanning unicyclic subgraph R . Thus using (2.7) to calculate $|x_C|^2$ we obtain

$$(2.8) \quad \tau |x_C|^2 = \sum_{\{A,B\}} \mathcal{F}_{\{A,B\}}(x_C).$$

On the other hand, x_F is orthogonal to every element of $\text{Cut}(G)$. Consequently, we have

$$\mathcal{F}_{\{A,B\}}(x) = \mathcal{F}_{\{A,B\}}(x_C)$$

for any two-component spanning forests $\{A, B\}$, allowing us to obtain the desired formula from (2.8). $\square$

2.3. Apex trees. If T is a weighted tree, then its vertex set V is a basis for $L(T)$. Let V^* denote the dual basis for $L(T)^*$, and let $v^* \in V^*$ denote the dual basis element dual to $v \in V$.

Lemma 2.2. *Let T be a weighted tree with no bad vertices. Let v be a vertex such that $w(v) > d(v)$. If $a \in L(T)^*$ satisfies $a \cdot v \neq 0$, then $|a|^2 \geq |v^*|^2$.*

Proof. Let G be the graph obtained by adding to T a single vertex v_∞ joined by $w(u) - d(u) \geq 0$ edges to each vertex $u \in T$. We call such a graph G an *apex tree* and v_∞ its *apex*. By construction, $L(T)$ is isomorphic to the cut lattice $\text{Cut}(G)$ via $v \longleftrightarrow \partial^*(v)$. By hypothesis, there is at least one edge e between v and v_∞ .

The projection of $e \in C_1(G)$ to $\text{Cut}(G)^* = L(T)^*$ is the dual element v^* . Its square-flux across a two-component spanning forest $\{A, B\}$ of G is 1, if $A \cup B \cup e$ is a spanning tree of G , and 0, if not. By the projection length formula Lemma 2.1, it follows that

$$(2.9) \quad |v^*|^2 = \tau_e / \tau$$

where τ_e counts the number of spanning trees for G containing e .

Select any $a \in L(T)^*$ such that $a \cdot v \neq 0$. Write $x_C \in \text{Cut}(G)^*$ for its image under the identification $\text{Cut}(G)^* = L(T)^*$. Because the restriction map from $C_1(G)$ to $\text{Cut}(G)^*$ surjects, there exists $x \in C_1(G)$ such that $x = x_C + x_F \in \text{Cut}(G)^* \oplus \text{Flow}(G)^*$. We aim to show that there are at least τ_e two-component spanning forests such that $\mathcal{F}_{\{A,B\}}(x) \neq 0$. Then another application of Lemma 2.1 yields $|a|^2 \geq |v^*|^2$.

To achieve our aim, we construct an injection ϕ from spanning trees containing e to two-component spanning forests $\{A, B\}$ for which $\mathcal{F}_{\{A,B\}}(x) \neq 0$. Let R be a spanning tree containing e . Let $R_1, \dots, R_k$ be the connected components of $R - v$. We will consider the two-component spanning forests of the form $\{R_j, R - R_j\}$. These are precisely the spanning forests obtained from R by deleting an edge incident to v . Since every vertex of $G - v$ is contained in precisely one R_j , (2.4) implies that we have the following linear relation among cut vectors:

$$\chi_{G-v,v} = \sum_{j=1}^k \chi_{R_j, R-R_j}.$$

Since $\chi_{G-v,v} \cdot x = a \cdot v \neq 0$, this implies that there exists an index ℓ such that $\chi_{R_\ell, R-R_\ell} \cdot x \neq 0$. We define $\phi(R) = \{R_\ell, R - R_\ell\}$ for such an ℓ and observe that this satisfies the desired property that $\mathcal{F}_{\{R_\ell, R-R_\ell\}}(x)$ is nonzero. It remains to verify that ϕ is injective. We will show how to reconstruct R from $\phi(R)$, using the fact that $\phi(R)$ is a two-component spanning forest obtained from R by deleting an edge incident to v . Let $\phi(R) = \{A, B\}$. If neither A nor B contains the edge e , then R , being a tree containing e , must take the form $R = A \cup B \cup e$. Otherwise w.l.o.g. we assume that e is contained in A . This implies that $v, v_\infty \in A$ and hence that B is a subtree of T not containing v . Moreover, there is a unique edge f between v and

a vertex of B , since T is a tree. In this case R must take the form $R = A \cup B \cup f$. Thus we have verified that ϕ is an injection. $\square$

Corollary 2.3. *Under the assumptions of Lemma 2.2, the element $v^* \in L(T)^*$ is irreducible.*

Proof. Suppose that $v^* = a + b$ with $a, b \in L(T)^*$ both nonzero. Then $1 = v^* \cdot v = (a + b) \cdot v = a \cdot v + b \cdot v$. Hence we may assume w.l.o.g. that $a \cdot v \neq 0$. We have $|a|^2 \geq |v^*|^2$ by Lemma 2.2, while $|b|^2 > 0$, since $b \neq 0$. Then

$$|v^*|^2 = |a + b|^2 = |a|^2 + 2(a \cdot b) + |b|^2 > |v^*|^2 + 2(a \cdot b) + 0.$$

Therefore, $a \cdot b < 0$. This proves that v^* is irreducible. $\square$

Corollary 2.4. *Under the assumptions of Lemma 2.2, $0 < |v^*|^2 \leq 1$, and if $|v^*|^2 = 1$, then either $w(v) = 1$ or T contains a leaf $u \neq v$ with $w(u) = 1$.*

Proof. Equation (2.9) shows that $0 < |v^*|^2 \leq 1$, and $|v^*|^2 = 1 \iff \tau_e = \tau \iff$ every spanning tree of G uses $e \iff e$ is a cut-edge of $G \iff e$ is the sole edge of G incident with the apex vertex $\iff w(v) = d(v) + 1$ and $w(u) = d(u)$ for all $u \in V$, $u \neq v$. If $V = \{v\}$, then $w(v) = 1$; if not, then T contains a leaf $u \neq v$, and $w(u) = d(u) = 1$. $\square$

2.4. Reducible vertices. Let T be a weighted tree such that $L(T)$ is positive definite. Let v be a vertex in T , and let $T_1, \dots, T_k$ be the weighted trees obtained by deleting v from T . For each tree T_i , let $v_i \in T_i$ be the vertex adjacent to v . Let $\lambda_i \in L(T_i)^*$ denote the dual to v_i with respect to the vertex basis $V(T_i)$. We regard $\lambda_i \in L(T) \otimes \mathbb{Q}$ under the inclusion $L(T_i)^* \subset L(T_i) \otimes \mathbb{Q} \subset L(T) \otimes \mathbb{Q}$. Equivalently, $\lambda_i \in L(T) \otimes \mathbb{Q}$ is the orthogonal projection to $L(T_i) \otimes \mathbb{Q}$ of the element $v_i^* \in L(T)^*$ dual to v_i with respect to the vertex basis $V(T)$.

Form the sublattice $L_1 = L(T_1) \oplus \dots \oplus L(T_k) \subset L(T)$. It is primitive, because the basis $V(T) \setminus \{v\}$ for it completes to the basis $V(T)$ of $L(T)$. Its orthogonal complement $L_1^\perp \subset L(T) \otimes \mathbb{Q}$ is a 1-dimensional rational vector space with a symmetric, bilinear form. Let $\pi : L(T) \rightarrow L_1^\perp$ denote the orthogonal projection, and let $L_0 = \pi(L(T))$ be the image of the projection. It is a rational lattice. Thus, $L(T)$ is contained in the orthogonal direct sum $L_0 \oplus L_1^* = L_0 \oplus L(T_1)^* \oplus \dots \oplus L(T_k)^*$.

Lemma 2.5. *The projection of v to L_0 is irreducible in L_0 .*

Proof. Every vertex of T besides v projects to 0 in L_0 . Therefore, $\pi(v)$ generates L_0 , hence is irreducible. $\square$

Lemma 2.6. *If v is reducible, then there exists an index $1 \leq i \leq k$ such that either*

- (i) λ_i is reducible in $L(T_i)^*$, or
- (ii) $\lambda_i \in L(T_i)$.

Proof. Because $v \cdot v_i = -1$ for all i and $v \cdot u = 0$ for every $u \in V(T_i)$, $u \neq v_i$, it follows that the projection of v to $L(T_i)^*$ is $-\lambda_i$ for all i . We therefore obtain the orthogonal decomposition

$$v = \pi(v) - \lambda_1 - \dots - \lambda_k.$$

If v is reducible, then we can decompose $v = x + y$ with nonzero $x, y \in L(T)$ satisfying $x \cdot y \geq 0$. These have orthogonal decompositions

$$x = \bar{x} + x_1 + \cdots + x_k \quad \text{and} \quad y = \bar{y} + y_1 + \cdots + y_k$$

with $\bar{x}, \bar{y} \in L_0$ and $x_i, y_i \in L(T_i)^*$ for $i = 1, \dots, k$. Thus, $\bar{x} + \bar{y} = \pi(v)$ and $x_i + y_i = -\lambda_i$ for all i . Since $\pi(v)$ is irreducible in L_0 by Lemma 2.5, we have $\bar{x} \cdot \bar{y} \leq 0$, with equality only if $\bar{x} = 0$ or $\bar{y} = 0$. On the other hand,

$$x \cdot y = \bar{x} \cdot \bar{y} + \sum_{i=1}^k x_i \cdot y_i \geq 0.$$

If $\bar{x} \cdot \bar{y} < 0$, then there exists an index i such that $x_i \cdot y_i > 0$, so λ_i is reducible in $L(T_i)^*$, which returns conclusion (i). If instead no λ_i is reducible in $L(T_i)^*$, then $\bar{x} \cdot \bar{y} = 0$ and $x_i \cdot y_i = 0$ for all i . Hence $\{\bar{x}, \bar{y}\} = \{0, \pi(v)\}$ and $\{x_i, y_i\} = \{0, -\lambda_i\}$ for all i . Assume w.l.o.g. that $\bar{x} = 0$. Then x takes the form

$$x = - \sum_{i \in A} \lambda_i$$

for some $A \subseteq \{1, \dots, k\}$, and $A \neq \emptyset$, since $x \neq 0$. Because $L_1 \subset L(T)$ is primitive and $x \in L(T) \cap (L_1 \otimes \mathbb{Q})$, it follows that $x \in L_1$. Because the λ_i are drawn from orthogonal summands of L_1^* , it follows that $\lambda_i \in L(T_i)$ for all $i \in A \neq \emptyset$, which returns conclusion (ii). $\square$

3. FINALÉ

Proof of Theorem 1.1. We argue somewhat more: either the bad vertex v is irreducible, or else T contains a leaf of weight 1 (which is therefore irreducible).

Define trees T_i and vertices v_i as in Section 2.4. The tree T_i has no bad vertices by assumption. In addition, the degree of v_i in T_i is one less than its degree in T , which is $\leq w(v_i)$. Therefore, Corollary 2.3 implies that $\lambda_i \in L(T_i)^*$ is irreducible for all i .

If $\lambda_i \notin L(T_i)$ for all i , then Lemma 2.6 implies that v is irreducible, as desired.

If instead $\lambda_i \in L(T_i)$ for some i , then $|\lambda_i|^2 \in \mathbb{Z}$. By Corollary 2.4, we must have $|\lambda_i|^2 = 1$, and either $w(v_i) = 1$ or T_i contains a leaf $u \neq v_i$ with $w(u) = 1$. Either case yields a leaf of T of weight 1, as desired. $\square$

Example: Let T be the weighted tree with a single vertex v with $d(v) = 3$ and $w(v) = 2$ and three leaves of weights 1, 2, and 3. Then T contains a single bad vertex v , and $L(T)$ is positive definite (it is isometric to the lattice of integer points in Euclidean space), but v is reducible.

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